Ibrahim Johar Farooqi
BAI-6A
23K-0074
CN Lab - Lab 01 Tasks

In-Lab Tasks

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23K-0074
BAI-6A
CN - Lab1 - In-Lab Tasks

1-126	A
128-191	B
192-223	C
229-239	D
240-255	F

↳ Task 1 → address class identification

Address	Class
1) 10.250.1.1	A
2) 150.10.15.0	B
3) 192.14.2.0	C
4) 148.17.9.1	B
5) 193.42.1.1	C
6) 126.8.156.0	A
7) 220.200.23.1	C
8) 230.230.45.58	D
9) 177.100.18.4	B
10) 119.18.45.0	A
11) 249.240.80.78	E
12) 199.155.77.56	C
13) 117.89.56.45	A
14) 215.45.45.0	C
15) 199.200.15.0	C
16) 95.0.21.90	A
17) 33.0.0.0	A
18) 158.98.80.0	B
19) 219.21.56.0	C

↳ Default subnet masks (classful)

↳ class A : 255.0.0.0 → network = 1st octet

↳ class B : 255.255.0.0 → network = first 2 octets

↳ class C : 255.255.255.0 → network = first 3 octets

day / date:

↳ Task 2 → host & network identification

↳ circle Network portion

1) 177.100.18.4
2) 119.18.45.0
3) 209.240.80.78
4) 199.155.77.56
5) 117.89.56.45
6) 215.45.45.0
7) 192.200.15.0
8) 95.0.21.90
9) 33.0.0.0
10) 158.98.80.0
11) 217.21.56.0
12) 10.250.1.1
13) 150.10.15.0
14) 192.14.2.0
15) 148.17.9.1
16) 193.42.1.1
17) 126.8.156.0
18) 220.200.231

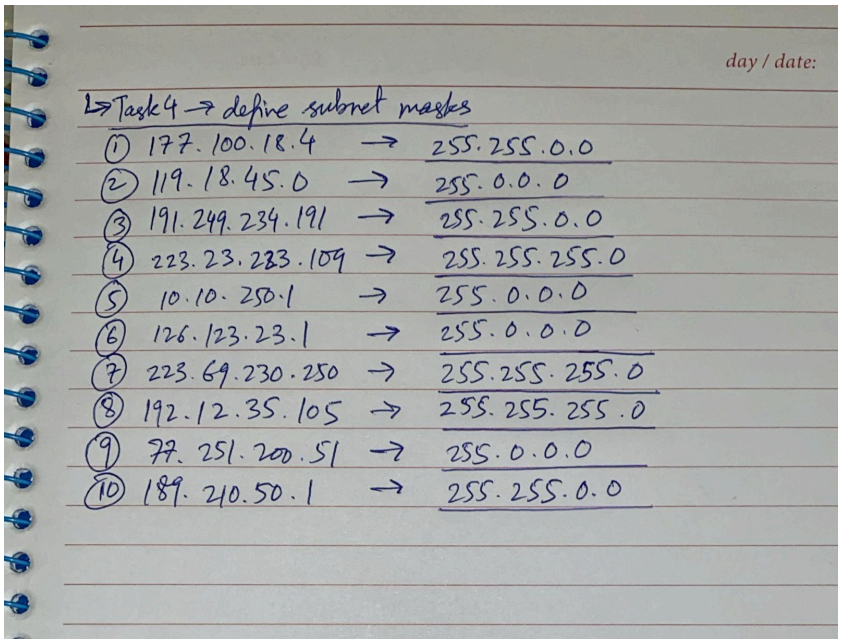
↳ circle Host portion

1) 10.15.123.50
2) 171.2.199.31
3) 198.125.87.177
4) 223.250.200.222
5) 17.45.222.45
6) 126.201.54.231
7) 191.41.35.112
8) 155.25.169.227
9) 192.15.155.2
10) 123.102.45.254
11) 148.17.9.155
12) 100.25.1.1
13) 195.0.21.98
14) 25.250.135.46
15) 171.102.77.77
16) 55.250.5.5
17) 218.155.230.14
18) 10.250.1.1

↳ Task 3 → Network Addresses (using subnet masks) & Host Addresses

Network Addresses	Host Addresses
1) 188.10.0.0	1) 0.0.18.2
2) 10.10.48.0	2) 0.0.0.80
3) 192.149.24.0	3) 0.0.0.11
4) 150.203.0.0	4) 0.0.230.19
5) 10.0.0.0	5) 0.10.10.10
6) 186.13.23.0	6) 0.0.0.11
7) 223.69.0.0	7) 0.0.23.20
8) 200.120.135.0	8) 0.0.0.215

KAGHAZ www.kaghaz.pk



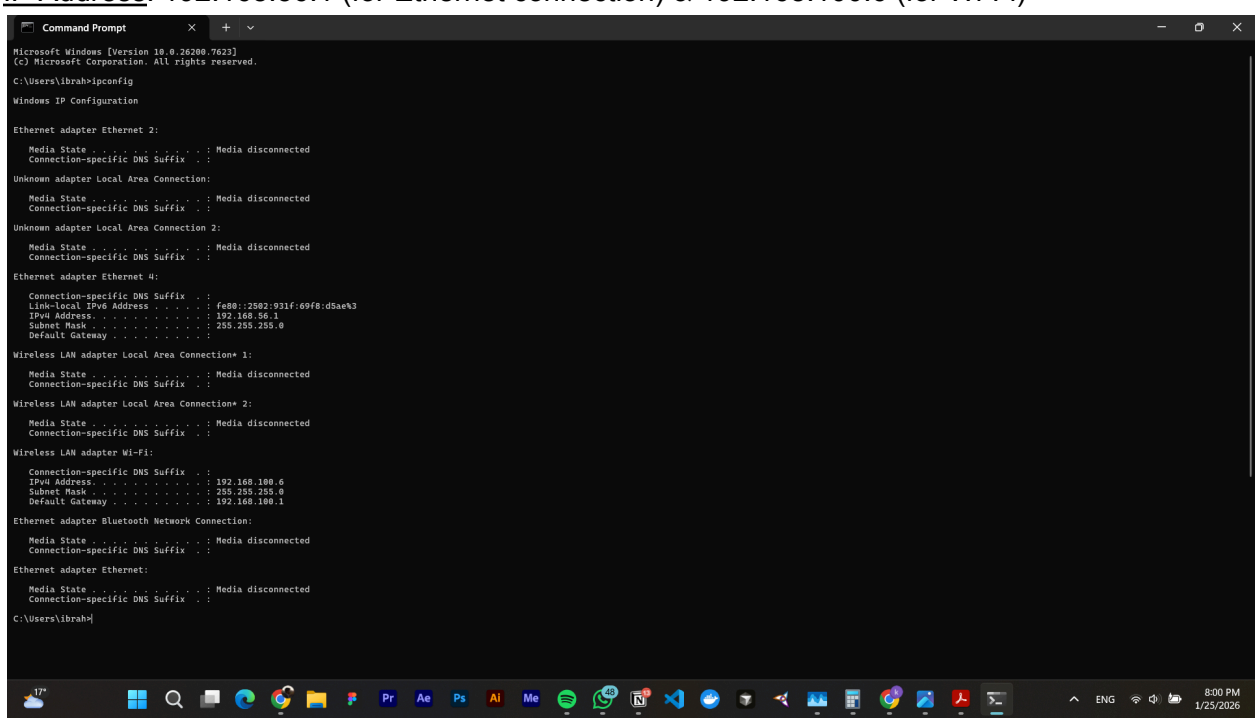
day / date:

↳ Task 4 → define subnet masks

1. 177.100.18.4 → 255.255.0.0
2. 119.18.45.0 → 255.0.0.0
3. 191.249.234.191 → 255.255.0.0
4. 223.23.233.109 → 255.255.255.0
5. 10.10.250.1 → 255.0.0.0
6. 126.123.23.1 → 255.0.0.0
7. 223.69.230.250 → 255.255.255.0
8. 192.12.35.105 → 255.255.255.0
9. 77.251.200.51 → 255.0.0.0
10. 189.240.50.1 → 255.255.0.0

Post-Lab Tasks

Task 1: Find the IP address of the computer you are currently using.
Command: ipconfig
IP Address: 192.168.56.1 (for Ethernet connection) & 192.168.100.6 (for Wi-Fi)



```
Command Prompt

Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ibrah>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Unknown adapter Local Area Connection 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 4:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2502:931f:69f8:d5ae%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 192.168.100.6
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.100.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

C:\Users\ibrah>
```

Task 2: Find the IP address of the computer you are currently using, plus MAC address, the gateway, plus whether DHCP is turned on.

Command: ipconfig /all

IP Address: 192.168.56.1 (for Ethernet connection) & 192.168.100.6 (for Wi-Fi)

MAC Address:

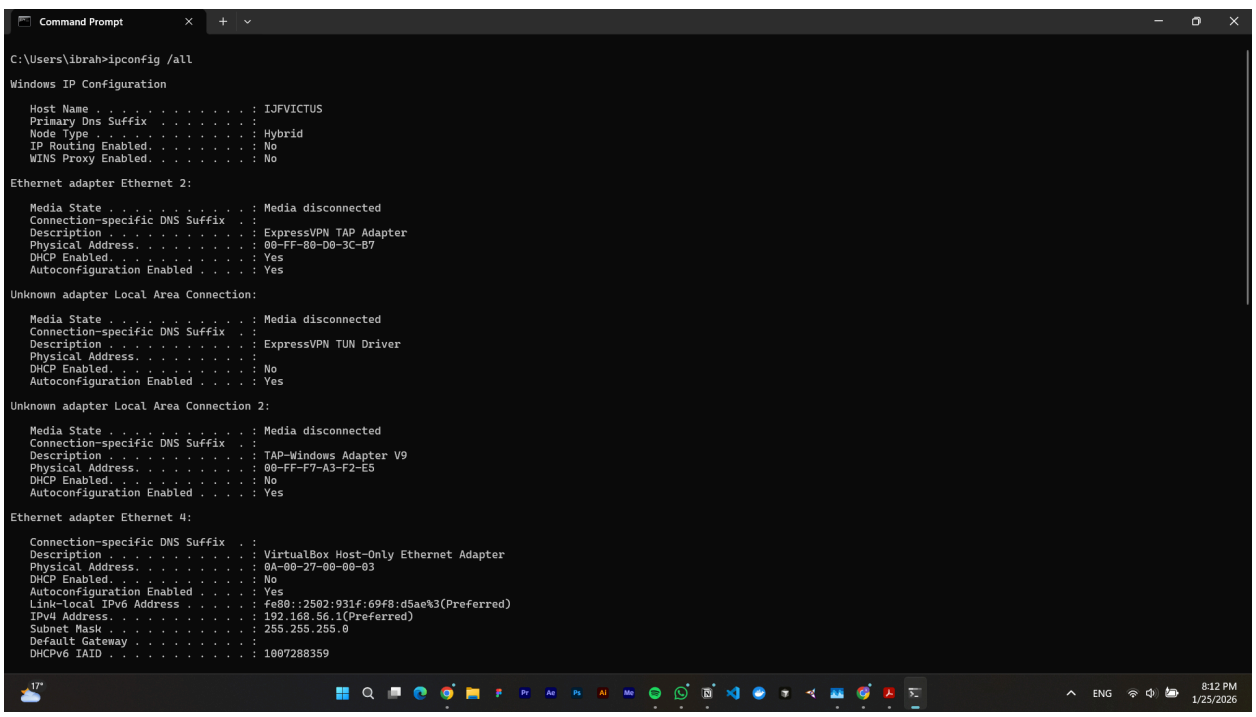
- *Ethernet*:
  - 0A-00-27-00-00-03
- *Wi-Fi:*
  - 14-13-33-8D-FA-C3

Gateway:

- *Ethernet*:
  - n/a
- *Wi-Fi:*
  - 192.168.100.1

DHCP:

- *Ethernet:* Not Enabled
- *Wi-Fi:* Enabled



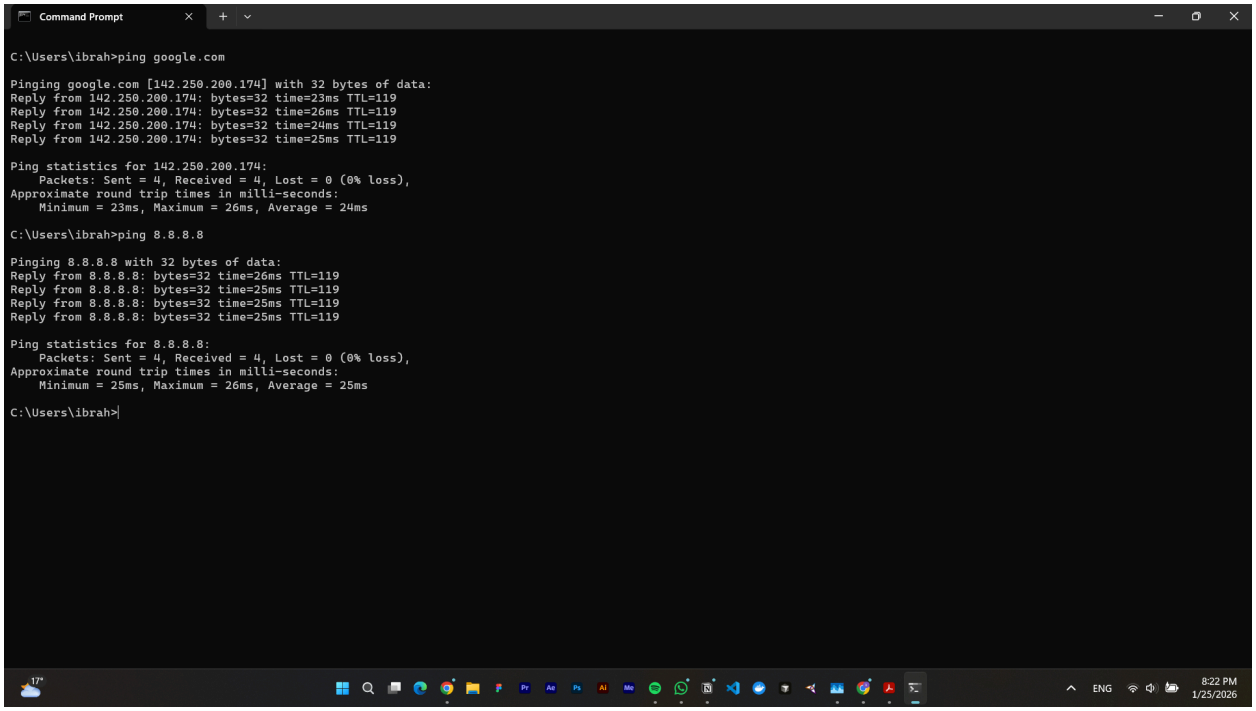
```
Command Prompt

C:\Users\ibrah>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : IJFVICTUS
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : ExpressVPN TAP Adapter
   Physical Address. . . . . . . . . : 00-FF-80-D0-3C-B7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Unknown adapter Local Area Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : ExpressVPN TUN Driver
   Physical Address. . . . . . . . . :
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Unknown adapter Local Area Connection 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : TAP-Windows Adapter V9
   Physical Address. . . . . . . . . : 00-FF-F7-A3-F2-E5
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 4:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-03
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2502:931f:69f8:d5ae%3(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 1007288359
```

```
Command Prompt

   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2A-12-72-16-84-69-93-58-63-FF
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 16-13-33-8D-FA-A3
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 16-13-33-8D-FA-B3
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 14-13-33-8D-FA-C3
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.100.6(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, January 25, 2026 10:43:26 AM
   Lease Expires . . . . . . . . . . : Monday, January 26, 2026 10:43:25 AM
   Default Gateway . . . . . . . . . : 192.168.100.1
   DHCP Server . . . . . . . . . . . : 192.168.100.1
   DNS Servers . . . . . . . . . . . : 192.168.100.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 14-13-33-8D-FA-C2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 84-69-93-58-63-FF
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\ibrah>
```

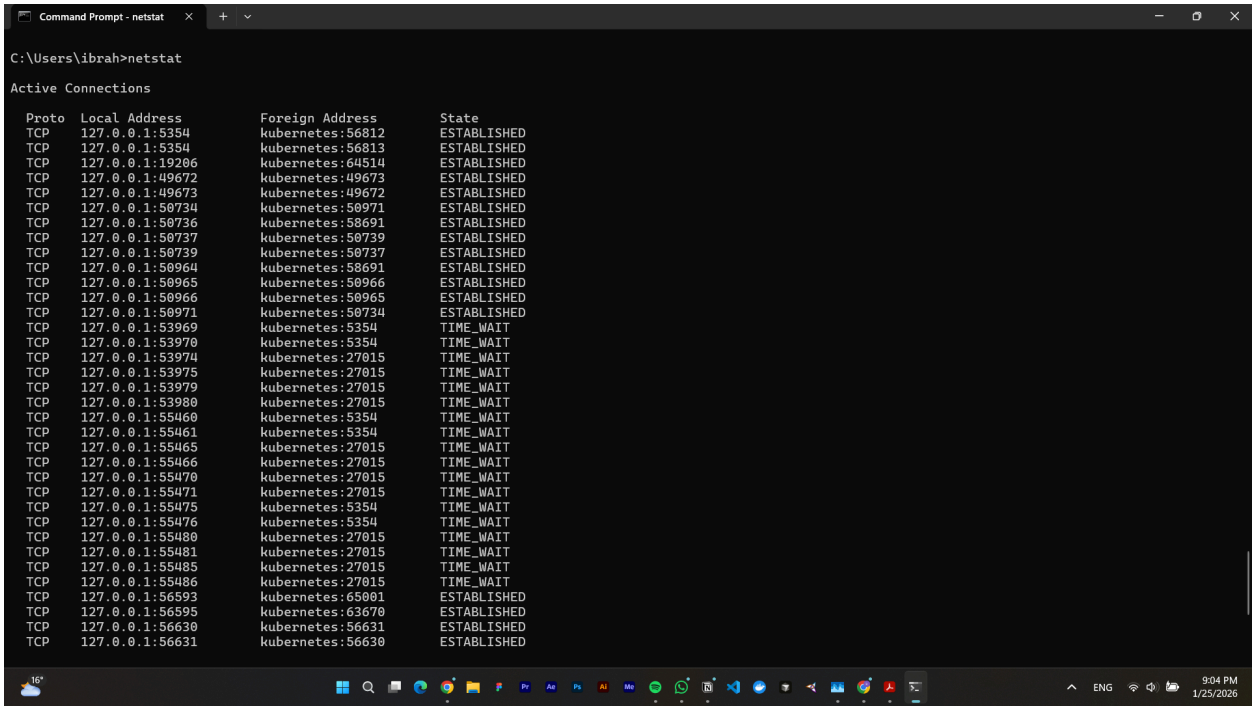
Task 3: Display the host name of the computer.
Command: hostname
Host Name: IJFVICTUS

```
Command Prompt

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 16-13-33-8D-FA-B3
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 14-13-33-8D-FA-C3
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv4 Address. . . . . . . . . . . : 192.168.100.6(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, January 25, 2026 10:43:26 AM
   Lease Expires . . . . . . . . . . : Monday, January 26, 2026 10:43:25 AM
   Default Gateway . . . . . . . . . : 192.168.100.1
   DHCP Server . . . . . . . . . . . : 192.168.100.1
   DNS Servers . . . . . . . . . . . : 192.168.100.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 14-13-33-8D-FA-C2
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek Gaming GbE Family Controller
   Physical Address. . . . . . . . . : 84-69-93-58-63-FF
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

C:\Users\ibrah>hostname
IJFVICTUS

C:\Users\ibrah>
```

Task 4: Check for basic IP connectivity between two computers by name and IP address. How can basic IP connectivity be checked? What are the reasons why there is no connectivity?
Command: ping
Reason: IP connectivity is checked with the “ping” command, which sends ICMP echo requests to the destination host and then waits for a reply from it. No connectivity may be caused by an incorrect IP address or hostname, the destination device being turned off, or even incorrect network configuration, like a gateway or subnet mismatch.

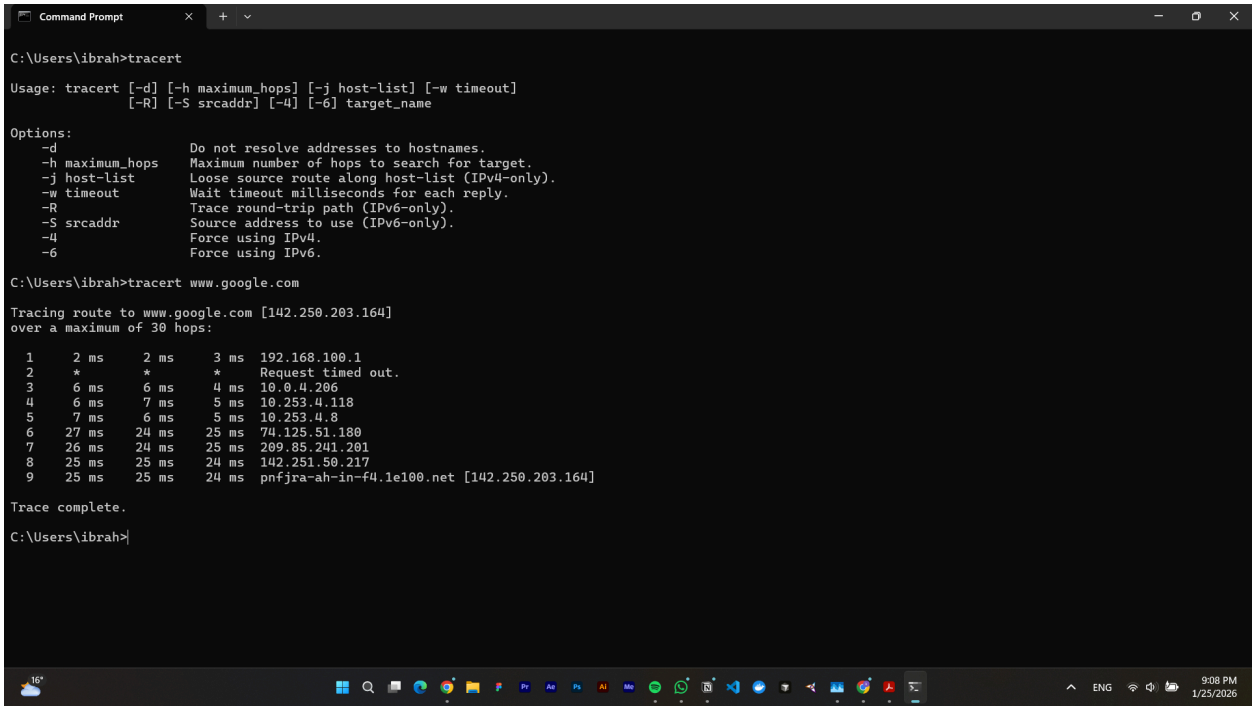


Task 5: Find out which ports on your host are connected to applications. Connect the browser to some external web page before running the appropriate command.
Command: netstat



```
Command Prompt - netstat

C:\Users\ibrah>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:5354         kubernetes:56812       ESTABLISHED
  TCP    127.0.0.1:5354         kubernetes:56813       ESTABLISHED
  TCP    127.0.0.1:19206        kubernetes:64514       ESTABLISHED
  TCP    127.0.0.1:49672        kubernetes:49673       ESTABLISHED
  TCP    127.0.0.1:49673        kubernetes:49672       ESTABLISHED
  TCP    127.0.0.1:50734        kubernetes:50971       ESTABLISHED
  TCP    127.0.0.1:50736        kubernetes:58691       ESTABLISHED
  TCP    127.0.0.1:50737        kubernetes:50739       ESTABLISHED
  TCP    127.0.0.1:50739        kubernetes:50737       ESTABLISHED
  TCP    127.0.0.1:50964        kubernetes:58691       ESTABLISHED
  TCP    127.0.0.1:50965        kubernetes:50966       ESTABLISHED
  TCP    127.0.0.1:50966        kubernetes:50965       ESTABLISHED
  TCP    127.0.0.1:50971        kubernetes:50734       ESTABLISHED
  TCP    127.0.0.1:53969        kubernetes:5354        TIME_WAIT
  TCP    127.0.0.1:53970        kubernetes:5354        TIME_WAIT
  TCP    127.0.0.1:53974        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:53975        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:53979        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:53980        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55460        kubernetes:5354        TIME_WAIT
  TCP    127.0.0.1:55461        kubernetes:5354        TIME_WAIT
  TCP    127.0.0.1:55465        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55466        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55470        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55471        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55475        kubernetes:5354        TIME_WAIT
  TCP    127.0.0.1:55476        kubernetes:5354        TIME_WAIT
  TCP    127.0.0.1:55480        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55481        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55485        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:55486        kubernetes:27015       TIME_WAIT
  TCP    127.0.0.1:56593        kubernetes:65001       ESTABLISHED
  TCP    127.0.0.1:56595        kubernetes:63670       ESTABLISHED
  TCP    127.0.0.1:56630        kubernetes:56631       ESTABLISHED
  TCP    127.0.0.1:56631        kubernetes:56630       ESTABLISHED
```

Task 6: Find the path of routers to www.google.com. What is its IP address? How many hops are involved in the path?
Command: tracert www.google.com
IP Address: 142.250.203.164
Total Hops: 9
Traceroute is used to find/trace the path taken by packets from the source to the destination by listing all of the intermediate routers (hops). Some hops may not be able to reply because of a firewall or security restriction.



```
C:\Users\ibrah>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.

C:\Users\ibrah>tracert www.google.com

Tracing route to www.google.com [142.250.203.164]
over a maximum of 30 hops:

  1     2 ms     2 ms     3 ms  192.168.100.1
  2     *        *        *     Request timed out.
  3     6 ms     6 ms     4 ms  10.0.4.206
  4     6 ms     7 ms     5 ms  10.253.4.118
  5     7 ms     6 ms     5 ms  10.253.4.8
  6    27 ms    24 ms    25 ms  74.125.51.180
  7    26 ms    24 ms    25 ms  209.85.241.201
  8    25 ms    25 ms    24 ms  142.251.50.217
  9    25 ms    25 ms    24 ms  pnfjra-ah-in-f4.1e100.net [142.250.203.164]

Trace complete.

C:\Users\ibrah>
```

Task 7: A ping to 192.168.0.2 works but a ping to the machine's name "blue machine" fails. What could be wrong?
Reason: If the ping to 192.168.0.2 works but the ping to the machine's name doesn't work, that's because of a DNS or name resolution problem. As the network connectivity seemingly is working due to the correct pinging to the IP address, but the hostname isn't being translated correctly to an IP address. This is because either the DNS server is not configured correctly, or the hostname is not registered in the DNS. Another possibility is that there is no corresponding entry in the local hosts file, causing this translation problem.

Task 8: Which type of cable will you use to connect in a normal home installation?
Answer: For a normal home installation, a straight-through Ethernet cable is used, as this type of cable is normally used to connect different devices, like a computer to a router.

Task 9: Can you connect a Switch to another Switch or a router to a PC using a straight-through cable? Explain your answer.
Answer: Historically, a straight-through cable couldn't be used to connect the same type of devices, such as a switch to a switch, but with modern networks, most devices now support Auto MDI/MDI-X, which allows the use of a straight-through cable for such connections, as the devices automatically adjust the transmitting and receiving pins.

Task 10: Write a brief report on your home network or any organizational network, including topology, 1 page max.

Answer: In terms of my home network, a typical home network is a local area network (LAN), which is used for everyday internet access and communication between multiple devices, such as phones and computers, within a limited geographical boundary of a house. The topology used in the home network can be categorised as a “Star Topology.” As all the devices in a home are connected to one central device, the Wi-Fi router. The Wi-Fi router acts as the main point through which all data is transmitted. If any device gets disconnected from the router, it doesn’t affect the rest of the network, which is why Star Topology is considered very reliable and easy to manage. The devices in this network comprise the wireless router, provided by the ISP (Internet Service Provider), as well as laptops, smartphones, and desktop computers. The Wi-Fi router performs several key functions, such as routing traffic, assigning IP addresses using DHCP, and providing wireless access. Devices are connected to the router via wireless connection or through an Ethernet cable for computers and laptops, which is usually a straight-through cable. The router assigns private IP addresses to all of the connected devices and forwards external traffic to and from the internet. The ISP provides DNS services, where it allows users to access websites using domain names instead of just IP addresses. All in all, this network layout provides a very simple but efficient home network that allows regular activities like web browsing and video streaming, etc.